

Tristan Misko

📍 Oakland, CA | ✉ tjmisko@quasiopitmal.io | [in linkedin.com/in/tristanmisko](https://www.linkedin.com/in/tristanmisko) | github.com/tjmisko | 🌐 tristanmisko.com

SUMMARY

I build software tools—data pipelines, programs, and AI applications—that automate the mechanical parts of computer work so people can attend to the parts of their work they care about most. Throughout 2026 I’ve been working on independent projects focused on integrating AI agents productively while bounding their behavior, controlling their output, and improving their ergonomics. I’m looking to join a fast-moving team building the tools across the stack that will make AI productive, reliable, and safe.

SKILLS & TOOLS

Backend & Systems	REST APIs in Flask, Django, Axum; persistence in SQLite, MongoDB; concurrency & DAG workflows.
Infrastructure & DevOps	Docker; Git; GitHub Actions; CI/CD; NGINX; systemd; SSH; Linux Server Management.
Applied AI Systems	RAG; Evals; Monitoring, Provenance, & Auditability; HITL Gating; Multi-Agent Autonomous Loops.
Programming Languages	Fluency from years of writing Python, JavaScript, Bash, R, and Stata. Learning to think, write, and build in Rust. I can read, reason about, and manage agents in Go, TypeScript, and Lua.

CURRENT PROJECTS

Functionary

Apr 2026 – Present

AI workflows, arranged on a visual canvas, with parallel execution, scoped toolset, HITL gates, and auditable provenance built-in.

- Directed AI agent implementation of Rust backend around a parallel DAG executor: single-owner scheduling, canonical-order semaphores to prevent deadlock on shared LLM quotas, and persistence and rehydration of runtime state in SQLite at human-review gates.
- React Flow canvas for placing blocks and structuring flows, with agent integration for sketching, progressively formalizing, and refactoring.
- Built with multi-agent autonomous loops across sandboxed Docker containers, governed by a DAG task ledger (700+ tasks over 90+ phases).

Lysilogy

Aug 2026 – Present

Preprocessor for scientific papers to reduce friction and improve understanding when reading and exploring new literatures.

- PDF preprocessing toolkit and views in Rust + TypeScript/React to find what’s worth reading in new scientific literature: automated visual paper sectioning and summaries, user-calibrated technical glossary generation, figure extraction and description, reference fetching.
- Tuned prompts on evaluations against a bank of golden examples; RAG on papers sourced from the citation graph for literature context.

Switchboard

Feb 2026 – Present

Live agent status indicators: cross-harness support, visible from any window, with jump to focus at a single keypress or click.

- Observes /proc for session discovery and processes agent lifecycle hook events, with transcript scanning fallback, to derive agent status.
- Window-manager integration on X11 & Wayland, mapping agent pid to controlling tty to terminal pane to window.
- Prototyped in Bash/tmux; directed agents to write a robust MVP version as Go daemon with RwMutex-based architecture to tune desired behavior; now implementing a fast, strongly-typed, mutex-free, single-owner actor design in Rust.

EXPERIENCE

Institute for Research on Labor and Employment @ UC Berkeley

Jul 2024 – Dec 2025

Software & Data Engineer | Berkeley, CA

Initiated and led a transition from manual to programmatic data collection, enabling a 30× expansion in the index’s data coverage.

- Made the case for technical improvements to data handling and reproducibility for the team’s research goals to non-technical stakeholders.
- Designed and built [sspi.world](#) for the Sustainable and Shared-Prosperity Index (SSPI): Flask & MongoDB backend running the five-stage ETL pipeline (collect, clean, compute, impute, finalize), a frontend in JavaScript with Chart.js, plus the CI/CD and Linux VPS underneath.
- Built a Click CLI for structuring access to HTTP endpoints, improving ergonomics for research apprentices and, in 2025, coding agents.

Berkeley Research Group

Aug 2023 – Jul 2024

Economics & Damages Associate | Emeryville, CA

Worked on data pipelines and qualitative research supporting antitrust litigation, and built out internal tooling to improve QC.

- Wrote Stata pipelines backing model inputs and figures in expert reports; code and output had to stand up to opposing expert scrutiny.
- Built BinderBuilder, a Python CLI parsing Word’s footnote XML from expert reports and auto-matching documents, then displaying them (using mupdf & fzf subprocesses) for human review. Per-footnote verification time fell from 60⁺s to 20s on the typical happy path, saving 2–4 hours per report.

Institute for Research on Labor and Employment @ UC Berkeley

Aug 2020 – Aug 2023

Research Apprentice → Research Team Lead | Berkeley, CA

Led two research team batches of 5–6 undergraduates in collection and analysis of economic and policy data. Began planning [sspi.world](#).

EDUCATION

University of California, Berkeley

Aug 2018 – May 2022

B.A. Applied Mathematics & B.A. Economics, Honors

+ Courses in Programming, Data Structures & Operating Systems (CS 61A/B/C); Machine Learning (CS 189, ECON C142); Probability & Statistics (STAT 134/135)